

Kerem Yildirim

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Munich, Germany



EDUCATION

- **M.Sc. Informatics**

Technical University of Munich

Thesis Topic: Object detection and scene completion from a single RGB-D frame

Grade Average: 1.7

October 2019 - September 2022, Germany

- **B.Sc. Computer Science and Engineering**

Sabanci University

Thesis Topic: Plant Disease Identification

Grade Average: 1.4

September 2015- June 2019, Turkey

EXPERIENCE

- **Working Student - Isarsoft**

Developing large scale object tracking and detection systems to be used in transportation systems.

October 2021 - Present, Germany

- **Working Student - Research Neutron Source FRM II**

Developed a system for high speed image segmentation using C++ and CUDA, enabling the processing of large datasets.

October 2020 - October 2021, Germany

- **Computer Vision Intern - WARP**

Designed and implemented a 3D object detection, instance segmentation and tracking system on LiDAR point clouds using Python and ROS to be used in an autonomous vehicle for in-campus transportation.

April 2020 - September 2020, Germany

- **Research Intern - University of Surrey**

Worked at CVSSP (Center of Vision, Sound and Signal Processing) and developed an ensemble of weak learners for plant classification. Achieved comparable results to state of the art deep learning methods.

June 2018 - August 2018, United Kingdom

LANGUAGES

- Turkish - Native
- English - C1
- German - B1

SKILLS

- C/C++, Python, CUDA, MATLAB, $\LaTeX$
- ROS, PyTorch, Tensorflow, OpenCV, OpenMP
- Linux, CMake, Git, Docker, AWS

PROJECTS

- **Monte Carlo Dropout for Object Detection on Point Clouds**

Improved state of the art object detection performance on point clouds by leveraging uncertainty.

Python, PyTorch

- **Indirect Visual Inertial Odometry with Optical Flow**

Extended the classical odometry pipeline by eliminating the descriptor matching and integrating frame to frame optical flow for more robust feature tracking.

C++, CMake, Eigen, Sophus, Ceres

- **KinectFusion**

Implemented KinectFusion algorithm by Newcombe et al. for real-time reconstruction of real world 3D scenes from depth images.

CUDA, C++, CMake, Eigen, Ceres

- **Divergence-Free Shape Correspondences**

Built a system for estimating non-rigid deformation of point cloud sequences without using any learning based approaches.

Python

PUBLICATIONS

- F. Tasyaran, K. Yildirim, K. Kaya, and M. K. Tas, "One table to count them all: Parallel frequency estimation on single-board computers," in *Euro-Par*, 2019
- S. Atito, B. A. Yanikoglu, E. Aptoula, I. Ganiyusufoglu, A. Yildiz, K. Yildirim, B. Sevilmis, and M. U. Sen, "Plant identification with deep learning ensembles," in *CLEF*, 2018